

**Re-evaluating a Standard Blood-Brain
Barrier Permeability Benchmark:
Representation, Chemical Novelty, and
Label Quality**

Ryan Yeung

Contents Page

Abstract.....	3
1. Introduction.....	4
2. Methods.....	7
2.1 Dataset.....	7
2.2 Molecular representations.....	8
2.3 Data splitting.....	8
2.4 Models.....	9
2.5 Evaluation.....	9
2.6 Similarity stratification.....	10
2.7 Label quality.....	10
2.8 Software and reproducibility.....	10
2.9 Supplementary analyses.....	11
3. Results.....	11
3.1 Physicochemical properties greatly differ between classes.....	11
3.2 Fingerprints outperform four descriptors.....	12
3.3 Descriptor coefficients favour polarity terms.....	13
3.4 Performance degrades monotonically with chemical novelty.....	13
3.5 Approximately 1% of labels are self-contradictory.....	16
3.6 The four descriptors are substantially collinear.....	18
3.7 Maximum Tanimoto similarity saturates for some structurally distinct pairs.....	20
4. Discussion.....	21
4.1 What the representation comparison means.....	21
4.2 The five-seed to twenty-seed shift.....	23
4.3 Chemical novelty and what headline figures conceal.....	24
4.4 Label contradictions.....	26
4.5 Limitations.....	28
4.6 Conclusion.....	30
5. Acknowledgements.....	32
6. References.....	33
Appendix A — Data availability.....	34

Abstract

Blood-brain barrier (BBB) penetration is a widely used ADMET prediction benchmark, with 12 of 25 published models on the Therapeutics Data Commons BBB_Martins leaderboard exceeding an AUROC of 0.90. This study evaluated 2,030 compounds by comparing logistic regression using four physicochemical descriptors with a Random Forest (RF) using 2,048-bit Morgan fingerprints across 20 scaffold splits, with paired statistical testing, similarity stratification and label inspection. The fingerprint model achieved a mean AUROC of 0.903 compared with 0.850 for the descriptor model, a difference of +0.053 (95% CI 0.038-0.068), while TPSA alone achieved 0.859 on a single split. Performance decreased monotonically with chemical novelty, from an AUROC of 0.965 in the most similar quartile to 0.773 in the least similar, a trend not explained by class balance. Ten structures, representing 1.0% of dataset rows, carried contradictory labels, arising partly from name-variant duplication and partly reflecting the limitations of a binary endpoint where brain exposure can depend on passive diffusion, active efflux and carrier-mediated uptake. Overall, a single pooled AUROC obscures both the predictive contribution of simple physicochemical properties and substantial variation with chemical novelty, supporting similarity-stratified reporting alongside overall performance.

1. Introduction

Failure in central nervous system (CNS) drug development is particularly costly because an active compound has limited therapeutic value if it cannot reach an effective concentration within the brain. The blood-brain barrier (BBB) restricts the entry of many circulating substances into the CNS, creating an important pharmacokinetic obstacle for drug development. Inadequate brain exposure may only become apparent after an already large investment, further contributing to high development cost and time. Predicting BBB penetration early can therefore help prioritise compounds before further resources are committed to experimental testing and development.

BBB penetration is associated with physicochemical properties including molecular weight (MW), lipophilicity, topological polar surface area (TPSA) and hydrogen-bond donor count (HBD), which influence passive diffusion through lipid membranes. CNS multiparameter optimisation (CNS MPO), for example, combines ClogP, ClogD, MW, TPSA, HBD and pKa to guide CNS compound design (Wager et al., 2010). Wager et al. found that higher CNS MPO scores were associated with favourable passive permeability and lower P-glycoprotein (P-gp) efflux liability, although CNS MPO was designed as a multiparameter design tool rather than specifically as a predictor of CNS penetration. Importantly, physicochemical properties alone cannot determine brain exposure. Across 11,303 compounds, Wager et al. found little correlation between lowering ClogP and reducing P-gp efflux liability. Net CNS exposure can also depend on P-gp-mediated efflux and carrier-mediated uptake, making the distinction between passive permeability and observed brain exposure important when defining a BBB prediction endpoint.

Machine-learning benchmarks allow molecular representations and algorithms to be compared systematically for absorption, distribution, metabolism, excretion and toxicity (ADMET) prediction. The Therapeutics Data Commons (TDC) includes BBB_Martins, a widely used binary BBB-penetration benchmark, where 12 of 25 published leaderboard entries exceed an AUROC of 0.90 (Koleiev et al., 2026). However, high leaderboard performance does not necessarily demonstrate model reliability. Koleiev et al. (2026), writing from Receptor.AI and declaring financial interests in the company, identified reproducibility, leakage and optimisation problems when auditing leading TDC ADMET models. MiniMol, ranked first on BBB_Martins, contained test compounds in its pretraining data; for example, danazol remained with unspecified stereochemistry despite appearing as a defined stereoisomer in the test set. CFA, ranked second, could not be reproduced because its code links were unavailable. Deliberate overfitting also moved an in-house model from rank 8 to rank 1 on BBB_Martins. These findings show why headline AUROC alone is insufficient for evaluating model reliability.

An overall AUROC can conceal how much predictive information comes from simple physicochemical properties, how performance changes with chemical novelty, which determines whether a model is useful for screening genuinely new compounds, and whether the BBB labels themselves are consistent. These are questions about evaluation rather than architecture and can be addressed by systematically examining a model's representation, test data and labels.

This study therefore addressed three questions:

1. How much additional predictive performance is gained from a 2,048-bit Morgan fingerprint compared with four physicochemical descriptors?
2. How does predictive performance change as test compounds become less similar to the training set?
3. What do contradictory labels within BBB_Martins reveal about the endpoint being predicted?

Logistic regression (LR) using MW, logP, TPSA and HBD was compared with a Random Forest using Morgan fingerprints across 20 scaffold splits, using paired statistical testing, similarity-stratified analysis and investigation of contradictory records.

2. Methods

2.1 Dataset

The BBB_Martins dataset was obtained through the Therapeutics Data Commons (TDC) Python interface (PyTDC v0.4.1), accessed via the `ADME` single-instance prediction class. The dataset has 2,030 rows describing 1,975 unique molecular structures, each represented as a SMILES string and labelled binary for blood-brain barrier penetration.

Class distribution was noticeably imbalanced, where 1,551 molecules (76.4%) were labelled as penetrant and 479 (23.6%) as non-penetrant. All 2,030 SMILES strings were parsed successfully by RDKit with no failures.

2.2 Molecular representations

Two representations were compared. The first being Morgan fingerprints, where circular fingerprints were generated using RDKit's

`rdFingerprintGenerator.GetMorganGenerator` with radius 2 and 2,048 bits, approximately equivalent to ECFP4. Each molecule is encoded as a binary vector indicating the presence or absence of hashed local atomic environments. The other representations used are physicochemical descriptors, where the following four descriptors were computed with RDKit: molecular weight, calculated logP (Wildman-Crippen), topological polar surface area, and number of hydrogen-bond donors. These were selected a priori as the canonical determinants of passive blood-brain barrier permeability rather than by feature selection on the data.

Features were computed once per unique structure and cached, so that identical molecules received identical representations across all splits.

2.3 Data splitting

Splits were generated using TDC's `get_split(method='scaffold')`, which computes a Bemis-Murcko scaffold for each molecule, groups molecules by scaffold, shuffles the groups under a given random seed, and assigns whole scaffold groups to training, validation and test partitions in a 70:10:20 ratio. Because assignment is by group, no scaffold appears on both sides of the split. Each split yielded 1,421 training and 406 test molecules.

The dataset contains 1,025 unique scaffolds across 2,030 molecules. The most frequent were benzene (136 molecules), the empty scaffold corresponding to acyclic compounds (99), and a steroid nucleus (76).

Twenty independent splits (seeds 1-20) were generated. Both models were fitted to the same split at each seed, allowing for paired statistical comparison.

This procedure is different from the official TDC Benchmark Group protocol, where the test partition is fixed and only the training and validation sets are reshuffled across runs. Resampling the entire split at each seed means the AUROC values reported here are not directly comparable to leaderboard entries, so the trade-off is deliberate so that sensitivity to scaffold assignment can be quantified, which a fixed test partition cannot measure.

2.4 Models

The fingerprint model was a random forest classifier (scikit-learn), 500 trees, `class_weight='balanced'` to compensate for class imbalance, and all other parameters at library defaults. No hyperparameter tuning was performed.

The descriptor model was a logistic regression with `class_weight='balanced'`, preceded by standardisation within a scikit-learn pipeline so that scaling parameters were fitted on training data only and not leaked from the test partition.

2.5 Evaluation

Area under the receiver operating characteristic curve (AUROC) was the primary metric, calculated from predicted class probabilities. AUROC was selected over accuracy because the majority-class rate of 76.4% renders accuracy uninformative, and over AUPRC because the positive class is the majority here, placing the AUPRC chance baseline at 0.764.

Models were compared by paired t-test and Wilcoxon signed-rank test on seed-wise differences, with a 95% confidence interval on the mean difference and Cohen's d_z as an effect size.

2.6 Similarity stratification

For every test molecule, Tanimoto similarity was computed against all training molecules using the same fingerprints, and the maximum retained as a measure of distance from the training set. Predictions from all 20 seeds were pooled ($n = 8,120$) and divided into quartiles by this maximum similarity. AUROC was computed within each quartile, with 95% confidence intervals from 2,000 percentile bootstrap resamples.

2.7 Label quality

Structures appearing more than once were identified by exact SMILES match, and those carrying more than one distinct label were flagged as contradictory.

2.8 Software and reproducibility

All analyses were performed in Python 3.11.16 using PyTDC 0.4.1, RDKit 2026.03.6, scikit-learn 1.9.0, SciPy 1.17.1, pandas 3.0.5, NumPy 2.4.6 and matplotlib 3.11.1. Analyses were run on a consumer laptop using CPU only, with a total runtime of under 30 minutes across all four scripts.

Scripts, per-seed results, per-molecule predictions and figures are available at github.com/RyanYeung111/bbb-permeability-benchmark, with environment specifications in environment.yml and requirements.txt.

2.9 Supplementary analyses

Pearson correlations were calculated pairwise across the four physicochemical descriptors using the full dataset. Variance inflation factors (VIFs) were calculated as $1/(1-R^2)$, where R^2 was obtained by regressing each descriptor against the remaining three using ordinary least squares. A VIF of 1 indicates no collinearity, while values above 5 are conventionally considered problematic and values above 10 indicate severe collinearity.

Logistic regression models were fitted separately using MW, logP, TPSA and HBD, with the same standardisation pipeline and class weighting as the four-descriptor model. This analysis was performed on scaffold split seed 1 only and was therefore intended as an indicative rather than definitive comparison.

For each test molecule, the training molecule with the highest Tanimoto similarity was recorded together with its similarity value. Pairs with similarity ≥ 0.99 were extracted and their Bemis-Murcko scaffolds compared to determine whether near-identical fingerprint similarity reflected failure of scaffold separation or limitations of the Morgan fingerprint representation.

These supplementary analyses were implemented in [bbb_step4.py](#).

3. Results

3.1 Physicochemical properties greatly differ between classes

Mean descriptor values separated strongly by class, in the directions predicted by passive diffusion theory.

Class	MW	log P	TPSA (Å²)	HBD
Non-penetrant (n = 479)	437.8	1.17	126.5	3.16
Penetrant (n = 1,551)	316.1	2.67	53.7	1.05

Table 1 (Above), Mean descriptor values by class

Penetrant molecules were smaller, more lipophilic, less polar and possessed fewer hydrogen-bond donors. Topological polar surface area differed by a factor of 2.4 between classes.

3.2 Fingerprints outperform four descriptors

Across 20 scaffold splits, the fingerprint random forest achieved a mean AUROC of 0.903 (SD 0.026) and the four-descriptor logistic regression 0.850 (SD 0.035).

Comparison	Value
Mean paired difference	+0.053
95% confidence interval	+0.038 to +0.068
Paired <i>t</i> -test	$t = 7.53, p < 0.0001$
Wilcoxon signed-rank	$p < 0.0001$
Effect size <i>dz</i>	1.68

Table 2 (Above), Paired model comparison statistics

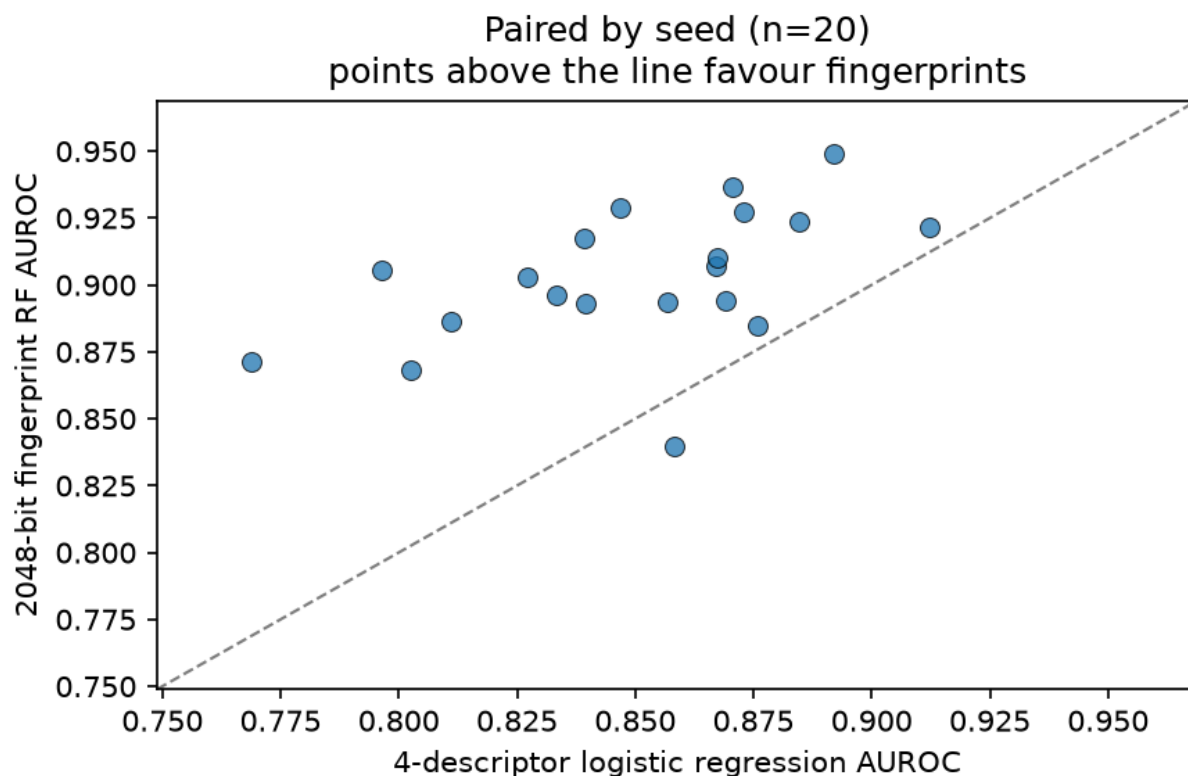


Figure 1 (Above), Paired per-seed AUROC for the fingerprint random forest versus the four-descriptor logistic regression across 20 scaffold splits. Points above the diagonal favour the fingerprint model.

The confidence interval excluded zero, establishing the fingerprint advantage as statistically robust. The fingerprint model outperformed the descriptor model on 19 of 20 splits; on seed 16 the ordering reversed (0.840 vs 0.858).

An earlier analysis restricted to five seeds produced a mean difference of +0.047 with a 95% confidence interval of -0.010 to +0.103 ($p = 0.083$), and was therefore unable to distinguish the two models. The effect was present but the analysis was underpowered.

3.3 Descriptor coefficients favour polarity terms

Standardised logistic regression coefficients, averaged across 20 seeds:

Descriptor	Coefficient	SD
TPSA	-0.814	0.147
HBD	-0.807	0.119
MW	-0.222	0.141
logP	+0.190	0.091

Table 3 (Above), Standardised logistic regression coefficients

These coefficients should be interpreted alongside the collinearity analysis reported in 3.6.

3.4 Performance degrades monotonically with chemical novelty

Pooling 8,120 test predictions across 20 seeds and stratifying by maximum Tanimoto similarity to the training set:

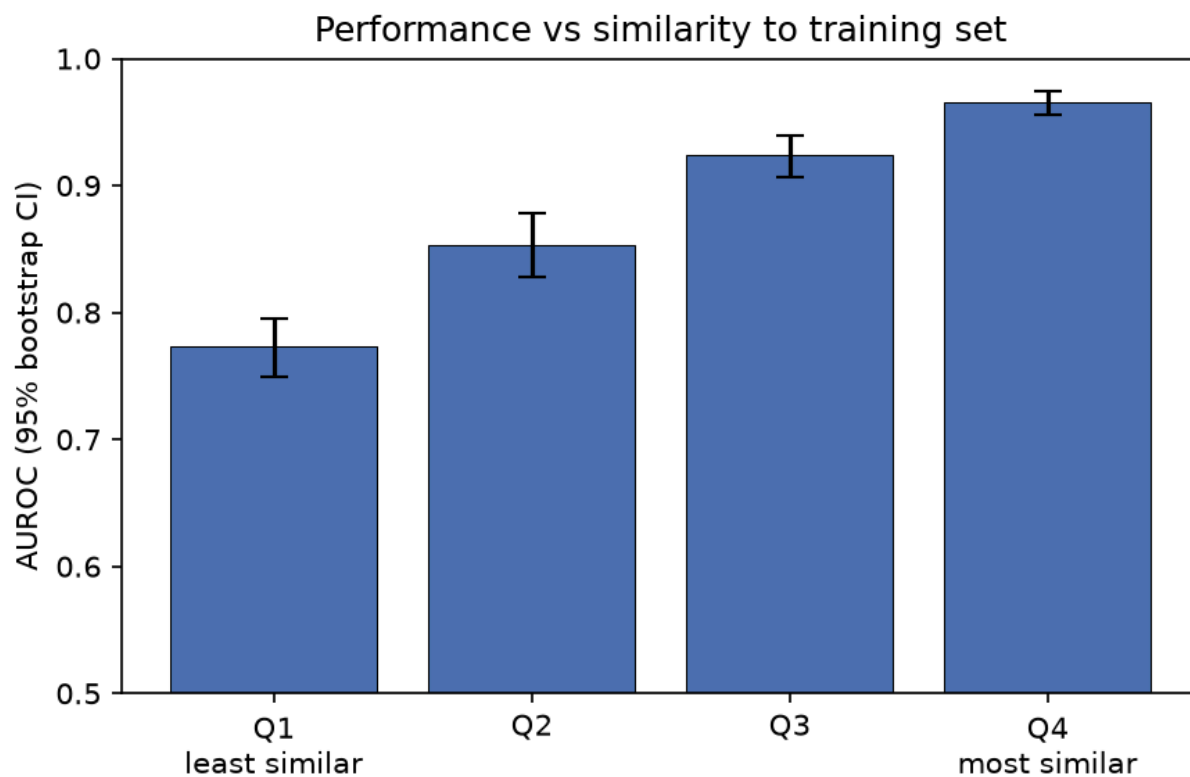
Quartile	Similarity range	n	Positive rate	AUROC	95% CI
Q1 (least similar)	0.040-0.313	2,033	0.78	0.773	0.749-0.795
Q2	0.313-0.446	2,037	0.82	0.853	0.828-0.878
Q3	0.446-0.595	2,037	0.78	0.924	0.907-0.939
Q4 (most similar)	0.595-1.000	2,013	0.70	0.965	0.955-0.974

Table 4 (Above), AUROC by similarity quartile

Performance increased monotonically across quartiles, spanning 0.192 AUROC. Confidence intervals for adjacent quartiles did not overlap at any point.

Class balance does not account for this gradient: the positive rate was non-monotonic across quartiles (0.78, 0.82, 0.78, 0.70), and the highest performing quartile had the lowest positive class prevalence.

Figure 2 (Below), AUROC by maximum-Tanimoto-similarity quartile, pooled across 20 scaffold splits (n = 8,120 predictions). Error bars show 95% bootstrap confidence intervals.



Median maximum similarity across seeds was 0.448, indicating that scaffold splitting separates ring systems while leaving substantial functional-group overlap between partitions.

3.5 Approximately 1% of labels are self-contradictory

Fifty-five duplicated SMILES entries were identified. Ten unique structures, accounting for 20 rows (1.0% of the dataset), carried both labels.

Structure	Entry identifiers
Acetylsalicylic acid	acetylsalicylate / aspirin
Trimetrexate	Trimetrexate / trimetrexate
Indomethacin	indomethacin / indomethacin(indometacin)
Methylprednisolone	Methylprednisolone / methylprednisolone
Miconazole	Miconazole / miconazole
Atropine	atropine / atropine(hyoscyamine)
Loperamide	BRL53080 / loperamide
Loratadine	loratadine (duplicated)
Levodopa	levodopa (duplicated)
Mequitazine	mequitazine (duplicated)

Table 5 (Above), Structures carrying contradictory labels

Six of the ten differ only by capitalisation or synonym, consistent with the merging of source lists without deduplication.

Since identical structures share a scaffold, contradictory duplicates are always assigned to the same split partition and cannot leak between training and test.

Loperamide appeared in the test partition in 6 of the 20 splits. Both models predicted it as BBB-penetrant on every occasion: the Random Forest gave a mean probability of 0.883 (range 0.796-0.922), compared with 0.731 (range 0.701-0.763) for logistic regression. Because the two entries for this structure carry opposite labels but receive identical predictions, each model necessarily classified exactly one correctly and one incorrectly on every appearance.

3.6 The four descriptors are substantially collinear

Pairwise Pearson correlations showed substantial relationships between several of the four physicochemical descriptors.

	MW	logP	TPSA	HBD
MW	1.000	0.134	0.705	0.514
logP	0.134	1.000	-0.482	-0.494
TPSA	0.705	-0.482	1.000	0.857
HBD	0.514	-0.494	0.857	1.000

Table 6 (Above), Pearson correlation matrix

The strongest relationship was between TPSA and HBD ($r = 0.857$), as hydrogen-bond donors contribute directly to polar surface area; MW and TPSA were also strongly correlated ($r = 0.705$).

Descriptor	R ²	VIF
TPSA	0.901	10.09
MW	0.796	4.90
HBD	0.751	4.01
logP	0.679	3.12

Table 7 (Above), Variance inflation factors

TPSA exceeded the threshold for severe collinearity, with 90.1% of its variance explained by the other three descriptors, while the remaining descriptors had VIF values below 5.

Model	AUROC (seed 1)
TPSA alone	0.859
HBD alone	0.839
logP alone	0.730
MW alone	0.702
All four descriptors	0.869
Fingerprint random forest	0.894

Table 8 (Above), Single-descriptor model performance

On this scaffold split, TPSA alone achieved an AUROC 0.010 below the four-descriptor model and 0.035 below the fingerprint random forest.

3.7 Maximum Tanimoto similarity saturates for some structurally distinct pairs

Of the 8,120 test predictions, 18 (0.22%) had a training neighbour with a Tanimoto similarity ≥ 0.99 . These cases corresponded to two distinct pairs of molecules, each appearing in both test-train orientations across splits. In both pairs, the molecules had different Bemis-Murcko scaffolds and were therefore correctly assigned to different partitions.

Both pairs had a Tanimoto similarity of 1.000. Cyclopropane (C1CC1) and cyclohexane (C1CCCCC1) produced identical radius-2 Morgan fingerprints because every atom in both structures is a methylene bonded to two other methylenes, so no radius-2 atomic environment distinguishes the three-membered from the six-membered ring. Similarly, trihexyphenidyl and procyclidine differ in their saturated nitrogen heterocycle, containing piperidine and pyrrolidine rings respectively, but this difference in ring size was not resolved by the radius-2 fingerprint.

These cases therefore reflect the resolution limit of the radius-2 Morgan fingerprint rather than failure of scaffold separation.

4. Discussion

4.1 What the representation comparison means

The fingerprint Random Forest has a mean AUROC of 0.903 (SD 0.026) while the four descriptor logistic regression had a mean AUROC of 0.850 (SD 0.035). The mean paired difference was +0.053 (95% CI +0.038 to +0.068, $p < 0.0001$, $d_z = 1.68$), with the random forest outperforming logistic regression in 19 out of 20 scaffold splits. This falls within the reported high-performance range for BBB_Martins, where 12 of 25 leaderboard models exceed an AUROC of 0.90 (Koleiev et al., 2026), however, the results are not directly comparable as leaderboard entries are evaluated on the TDC Benchmark Group's single fixed test partition, whereas the current study used repeated scaffold splits. Importantly, Koleiev et al. also advises that AUROC values above 0.90 on these benchmarks should not, by themselves, be interpreted as evidence of overfitting or test set leakage.

The size of the improvement is more informative than whether it is statistically significant. The confidence interval puts the random forest advantage at around 0.04-0.07 AUROC. The logistic regression used only molecular weight, logP, topological polar surface area and hydrogen-bond donors, while the random forest used a 2,048-bit Morgan fingerprint encoding much more structural information. The four-descriptor model accounted for about 87% of the random forest's improvement over chance (AUROC = 0.5). These results suggest that for this dataset and endpoint, much of the predictive signal can be captured by simple physicochemical properties.

Morgan fingerprints also retain local substructural information that the four aggregate descriptors do not hold, including specific ring systems and functional group combinations, which may give additional information associated with BBB penetration. Both models were both

tested using library defaults, providing an equal comparison rather than optimisation favouring either model.

The LR coefficients were largest for topological polar surface area (-0.814) and hydrogen bond donors (-0.807), compared with molecular weight (-0.222) and logP ($+0.190$), which would suggest a stronger contribution from polarity and hydrogen bonding properties. However, these coefficients cannot be interpreted as independent measures of importance because the predictors were collinear. Topological polar surface area had a VIF of 10.09 and correlated strongly with hydrogen bonding ($r = 0.857$). Single descriptor models nevertheless provided supporting evidence, with topological polar surface area achieving an AUROC of 0.859 by itself and 0.839 for hydrogen bond donors, compared with 0.730 for logP and 0.702 for molecular weight. On the same single scaffold split, topological polar surface area alone was only 0.010 below the four-descriptor model and 0.035 below the fingerprint random forest. However, these single split results should be considered indicative until confirmed across repeated splits.

4.2 The five-seed to twenty-seed shift

The estimated performance difference between the two models was similar whether five or twenty scaffold splits were evaluated. Across five splits, the mean paired AUROC difference was +0.047 (95% CI -0.010 to +0.103; $p = 0.083$), compared with +0.053 (95% CI +0.038 to +0.068; $p < 0.0001$) across twenty splits. The estimated effect therefore changed by only 0.006, whereas the confidence interval narrowed substantially. Increasing the number of splits primarily improved the precision of the estimate rather than changing its magnitude.

The five-split analysis illustrates the distinction between failing to demonstrate a difference and demonstrating that no difference exists. Its confidence interval, from -0.010 to +0.103, was compatible with both a small advantage for LR and a substantially larger advantage for RF, so equivalence could not be concluded. This limitation is particularly clear for non-parametric testing: with only five paired observations, the smallest possible two-sided Wilcoxon signed-rank p -value is 0.0625. Conventional statistical significance at $p < 0.05$ is therefore unattainable regardless of how consistently one model outperforms the other, so a non-significant result at this sample size is weak evidence of anything.

The high split-to-split variation arises from the structure of scaffold splitting itself. In seed 16, the model ordering reversed, with RF achieving an AUROC of 0.840 compared with 0.858 for LR. Scaffold groups are allocated as complete blocks, and BBB_Martins contains several large groups, including compounds with a benzene scaffold (136 molecules), acyclic compounds (99) and a steroid scaffold (76). With only 406 compounds in each test partition, a 76-compound scaffold represents almost one fifth of the test set. Placement of such groups can therefore materially alter both training and test composition. The RF's split-to-split SD of 0.026 is approximately half the mean model difference of 0.053, showing that partition variability is on

the same order as the effect being measured. The ordering reversed in 1 of the 20 splits, so a single-split evaluation would have shown LR outperforming RF in that partition.

These findings indicate that single-split or few-split evaluations may be unable to reliably distinguish models separated by only a few hundredths of AUROC. Kamuntavičius et al. (2025) similarly found that apparent test-set improvements in ADMET models frequently failed cross-validation hypothesis testing, supporting repeated evaluation with hypothesis testing as a more robust alternative to single hold-out assessment. This is particularly relevant to scaffold-based evaluation, where allocation of large scaffold groups also contributes to differences in test-set composition and chemical similarity between splits, with median nearest-training similarity ranging from 0.344 to 0.512 in the present study. Repeated paired evaluation across multiple scaffold splits, with confidence intervals reported alongside mean performance, therefore provides a more reliable basis for model comparison at relatively little additional computational cost.

4.3 Chemical novelty and what headline figures conceal

Scaffold splitting is intended to provide a more challenging evaluation than random splitting by separating compounds according to their molecular scaffolds. However, this does not guarantee that test compounds are chemically dissimilar to the training set. In this study, the median maximum Tanimoto similarity between each test compound and its nearest training compound was 0.448. Therefore, despite having different Bemis-Murcko scaffolds, many test compounds retained considerable structural similarity to molecules seen during training.

Model performance was strongly related to this similarity. Across 8,120 pooled predictions from 20 independent scaffold splits, compounds were divided into quartiles according to their maximum Tanimoto similarity to the training set (Table 4, Figure 2). AUROC increased

monotonically from 0.773 in the least-similar quartile (Q1; Tanimoto 0.040-0.313) to 0.965 in the most-similar quartile (Q4), a difference of 0.192, with no overlap between adjacent confidence intervals. The model therefore performs substantially better when test compounds resemble chemistry already represented in training. Similarity was calculated using the same Morgan fingerprint representation supplied to the model, so this analysis measures novelty as the model perceives it.

This relationship was not explained by class balance. The proportion of BBB-positive compounds across Q1-Q4 was 0.78, 0.82, 0.78 and 0.70, respectively, with Q4 achieving the highest AUROC despite having the lowest prevalence. If class balance were driving the AUROC gradient, prevalence would need to vary systematically with it, which it does not.

This has implications for prospective screening. The overall scaffold-split AUROC of 0.903 combines compounds with very different levels of structural novelty and therefore conceals substantial variation in performance. For screening a novel chemical series, the least-similar compounds provide a closer approximation to the intended use than the pooled test set. The Q1 AUROC of 0.773 provides a more relevant estimate of performance on chemistry further from the training set. The model still retains useful discrimination, but its performance is substantially lower than the headline figure suggests.

This corresponds to the QSAR concept of an applicability domain, defined as the chemical and response space within which a model makes predictions with a given reliability (Netzeva et al., 2005). OECD Validation Principle 3 requires a defined applicability domain and notes that similarity to the training set can be used to assess whether a compound falls within it (OECD, 2007). The quartile analysis provides an empirical expression of this principle. ADMET benchmarks should therefore report performance by nearest-neighbour similarity alongside overall AUROC, rather than averaging across compounds of very different novelty.

4.4 Label contradictions

Ten structures in the BBB_Martins dataset have both positive and negative labels, accounting for 20 rows (1% of the dataset). As identical structures have identical Bemis-Murcko scaffolds, contradictory pairs remain in the same partition and therefore do not cause data leakages.

Although they technically limit maximum accuracy, their small number means the effect on performance is negligible. Instead, they show a limitation in the labelling scheme.

Six pairs differ only in their recorded drug names, e.g. Miconazole/miconazole, suggesting that datasets may have been merged by drug name rather than standardised structure.

Deduplication using canonical SMILES would have identified these conflicts at the point of merging.

Other contradictions reflect the complexity of BBB transport. Loperamide is highly lipophilic and can cross the BBB by passive diffusion, but P-glycoprotein actively pumps it back into the circulation. In *mdr1a*-knockout mice, brain exposure to loperamide and its metabolites increased approximately sevenfold, with a pilot toxicity study finding that oral loperamide acquired potent opiate-like central activity (Schinkel et al., 1996).

Levodopa presents the opposite case. As a polar zwitterion, its physicochemical properties would suggest poor passive permeability through the BBB. The 4F2hc/LAT1 complex participates in L-DOPA transport in an *in vitro* BBB model, and LAT1 was predominantly expressed in CNS microvessels, supporting a role in L-DOPA uptake *in vivo* (Kageyama et al., 2000). L-DOPA uptake was also strongly inhibited by large neutral amino acids, which explains why dietary protein can reduce levodopa efficacy.

Both models consistently predicted loperamide as BBB-penetrant, reproducing the prediction expected from its physicochemical properties alone. Although structural fingerprints may contain

features associated with transporter interactions, the fingerprint model did not recover the P-gp-limited exposure distinction in this case and was more confident than logistic regression (mean probability 0.883 vs 0.731).

Notably, the BBB-positive entry is recorded as BRL53080, whereas the BBB-negative entry is recorded as loperamide. Although the original sources of these individual labels cannot be established from the dataset alone, this naming difference is consistent with the mixed provenance of the endpoint.

Together, these examples show that brain exposure depends on passive diffusion, active efflux and carrier-mediated uptake. These mechanisms help explain the contradictions seen, as BBB_Martins labels originate from both thresholded logBB values and qualitative CNS-activity annotations, meaning different sources may assign conflicting labels to the same structure. Although structural fingerprints may capture some transporter-related features, both models are trained against a label that conflates passive permeability with net brain exposure. Therefore, the 0.053 AUROC improvement of fingerprints over the four descriptors should be interpreted in light of this limitation.

4.5 Limitations

This study only used one dataset and one endpoint, with no separate external validation set. Therefore, the results may not generalise to other ADMET datasets or different areas of chemical space. BBB_Martins also combines logBB values converted into BBB+ or BBB- labels with qualitative information about CNS activity. Compounds close to the logBB cut-off may therefore have uncertain classifications, separate from the contradictory duplicates discussed in 4.4.

In this study, duplicates were identified using exact SMILES without standardising the molecular structures first. In comparison, Kamuntavičius et al. (2025) standardised structures, including salts and tautomers, before checking for duplicates. With the same original dataset of 2,030 compounds, they removed 36 inconsistent duplicates and another 44 duplicates, while this study has only identified 10 contradictory structures. This may suggest that using exact SMILES may underestimate the number of duplicates in the dataset. Some multi-component structures also remained, such as propranolol recorded with a counterion ([Cl-]).

The repeated scaffold splitting protocol is different from the official TDC Benchmark Group evaluation, which uses a fixed test partition. Therefore, the AUROC values reported here should not be directly compared with leaderboard values. This protocol allowed split-to-split variability to be quantified, but that variability would still remain substantial, where the fingerprint Random Forest retained a standard deviation of 0.026 across 20 splits. Kamuntavičius et al. (2025) similarly acknowledge that their data cleaning makes their results technically incomparable with public leaderboard models, despite relatively few test-set compounds being removed.

Neither model underwent hyperparameter optimization, so the results represent equal baseline implementations rather than the maximum achievable performance of either architecture. Only

two representations were compared: logistic regression using four physicochemical descriptors and Random Forest using a 2,048-bit Morgan fingerprint. Graph neural networks and pretrained chemical language models were not evaluated. Kamuntavičius et al. reported that Random Forest underperformed relative to gradient-boosted models in their ADMET experiments and selected CatBoost for subsequent analyses, suggesting that a different learning algorithm could alter the observed +0.053 fingerprint advantage, plausibly widening it. Descriptor collinearity, particularly the TPSA VIF of 10.09, limits interpretation of individual logistic regression coefficients as discussed in 4.1, while the single-descriptor analysis was performed on only one scaffold split.

Chemical novelty was quantified using Tanimoto similarity calculated from the same radius-2 Morgan fingerprint supplied to the Random Forest. The analysis therefore measures novelty as perceived by the model rather than structural novelty in general. This representation can also saturate, as chemically distinct pairs such as cyclopropane/cyclohexane and trihexyphenidyl/procyclidine produced Tanimoto similarities of 1.000 because their fingerprints did not distinguish the relevant ring-size differences. This affected only 0.22% of predictions and does not indicate scaffold leakage, but demonstrates that maximum Tanimoto similarity is not equivalent to structural identity. Alternative applicability-domain measures, such as descriptor-space distance or leverage, were not evaluated. The collinearity and fingerprint-saturation limitations were identified through follow-up analyses rather than anticipated at the outset and are reported here for completeness.

4.6 Conclusion

Morgan fingerprints outperformed the four physicochemical descriptors by 0.053 AUROC (0.903 vs 0.850), a statistically robust but modest improvement given the substantially greater dimensionality of the fingerprint representation, while TPSA alone achieved an AUROC of 0.859 on a single split. Performance fell monotonically from 0.965 to 0.773 across similarity quartiles, a gradient not explained by class balance, showing that a pooled AUROC conceals substantial variation in model reliability across chemical space. Approximately 1% of dataset rows carried contradictory labels, arising partly from clerical duplication and partly from a binary endpoint that cannot represent transporter-mediated influences on brain exposure. ADMET benchmarks should therefore report performance stratified by similarity to the training set, because a single AUROC describes neither the contribution of simple physicochemistry nor performance on the novel chemistry these models are intended to screen.

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6. References

- Bemis, G. W.; Murcko, M. A. The Properties of Known Drugs. 1. Molecular Frameworks. *J. Med. Chem.* 1996, *39* (15), 2887-2893.
- Breiman, L. Random Forests. *Mach. Learn.* 2001, *45*, 5-32.
- Ertl, P.; Rohde, B.; Selzer, P. Fast Calculation of Molecular Polar Surface Area as a Sum of Fragment-Based Contributions and Its Application to the Prediction of Drug Transport Properties. *J. Med. Chem.* 2000, *43*, 3714-3717.
- Huang, K.; Fu, T.; Gao, W.; Zhao, Y.; Roohani, Y.; Leskovec, J.; Coley, C. W.; Xiao, C.; Sun, J.; Zitnik, M. Artificial Intelligence Foundation for Therapeutic Science. *Nat. Chem. Biol.* 2022, *18* (10), 1033-1036.
- Kageyama, T.; Nakamura, M.; Matsuo, A.; Yamasaki, Y.; Takakura, Y.; Hashida, M.; Kanai, Y.; Naito, M.; Tsuruo, T.; Minato, N.; Shimohama, S. The 4F2hc/LAT1 Complex Transports L-DOPA across the Blood-Brain Barrier. *Brain Res.* 2000, *879* (1-2), 115-121.
- Kamuntavičius, G.; Paquet, T.; Bastas, O.; Šalkauskas, D.; Prat, A.; Abdel Aty, H.; Pabrinkis, A.; Norvaišas, P.; Tal, R. Benchmarking ML in ADMET Predictions: The Practical Impact of Feature Representations in Ligand-Based Models. *J. Cheminform.* 2025, *17* (1), 108. DOI: 10.1186/s13321-025-01041-0
- Koleiev, I.; Stratiichuk, R.; Shevchuk, N.; Melnychenko, M.; Nyporko, A.; Todoryshyn, D.; Husak, V.; Starosyla, S.; Yesylevskyy, S.; Nafiiev, A. An End-User Audit of Reproducibility, Data Leakage, and Overfitting of the Top-Ranked ADMET Prediction Models in TDC Leaderboards. *J. Chem. Inf. Model.* 2026, *66* (14), 8045-8058. DOI: 10.1021/acs.jcim.6c00819

Martins, I. F.; Teixeira, A. L.; Pinheiro, L.; Falcão, A. O. A Bayesian Approach to in Silico Blood-Brain Barrier Penetration Modeling. *J. Chem. Inf. Model.* 2012, 52 (6), 1686-1697. DOI: 10.1021/ci300124c

Netzeva, T. I.; Worth, A.; Aldenberg, T.; et al. Current Status of Methods for Defining the Applicability Domain of (Quantitative) Structure-Activity Relationships: The Report and Recommendations of ECVAM Workshop 52. *Altern. Lab. Anim.* 2005, 33 (2), 155-173.

OECD. *Guidance Document on the Validation of (Quantitative) Structure-Activity Relationship [(Q)SAR] Models*; OECD Series on Testing and Assessment No. 69, ENV/JM/MONO(2007)2; OECD Publishing: Paris, 2007. DOI: 10.1787/9789264085442-en

Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; et al. Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* 2011, 12, 2825-2830.

RDKit: Open-Source Cheminformatics. <https://www.rdkit.org> (accessed September 7, 2026).

Rogers, D.; Hahn, M. Extended-Connectivity Fingerprints. *J. Chem. Inf. Model.* 2010, 50 (5), 742-754.

Schinkel, A. H.; Wagenaar, E.; Mol, C. A.; van Deemter, L. P-Glycoprotein in the Blood-Brain Barrier of Mice Influences the Brain Penetration and Pharmacological Activity of Many Drugs. *J. Clin. Invest.* 1996, 97 (11), 2517-2524. DOI: 10.1172/JCI118699

Wager, T. T.; Hou, X.; Verhoest, P. R.; Villalobos, A. Moving beyond Rules: The Development of a Central Nervous System Multiparameter Optimization (CNS MPO) Approach To Enable Alignment of Druglike Properties. *ACS Chem. Neurosci.* 2010, 1 (6), 435-449. DOI: 10.1021/cn100008c

Wildman, S. A.; Crippen, G. M. Prediction of Physicochemical Parameters by Atomic Contributions. *J. Chem. Inf. Comput. Sci.* 1999, 39, 868-873.

Appendix A — Data availability

All scripts, per-seed results, per-molecule predictions and figures are available at github.com/RyanYeung111/bbb-permeability-benchmark. The analysis can be reproduced end to end by creating the documented environment and running `bbb_step1.py` through `bbb_step4.py` in order.

Scripts

File	Contents
<code>bbb_step1.py</code>	Dataset loading, inspection and baseline random forest
<code>bbb_step2.py</code>	Label quality, descriptor model, median similarity split
<code>bbb_step3.py</code>	Twenty-split evaluation, paired testing, quartile stratification
<code>bbb_step4.py</code>	Descriptor collinearity, named per-molecule predictions, near-identical pairs

Generated files

File	Contents
<code>results_per_seed.csv</code>	AUROC for both models, per seed, with median similarity
<code>results_per_molecule.csv</code>	Every test prediction with label and nearest-neighbour similarity
<code>results_per_molecule_named.csv</code>	As above, with entry identifier and nearest-neighbour identity
<code>results_by_quartile.csv</code>	Quartile AUROCs with bootstrap intervals
<code>results_descriptor_collinearity.csv</code>	Per-descriptor R^2 and variance inflation factors
<code>results_similarity_anomalies.csv</code>	Test–train pairs with Tanimoto similarity ≥ 0.99

<code>fig_model_comparison.png</code>	Figure 1
<code>fig_similarity_quartiles.png</code>	Figure 2